

Supplementary Material S1

The 50 Refinement Rules of the Common Rule Engine

to accompany “Refine First, Trust Later: A Task-Relative Noise Refinement Framework and the Analysis of Five Illusions Noise Casts on Network Traffic Classification”

This supplement gives the full specification of the 50 refinement rules summarized in Section 3.3 and Table 3 of the article. The rules are grouped by the six categories of the noise taxonomy (Level 3), one section per category (Sections S1.2–S1.7); within the label-error category Bb, the rules are further grouped into common rules and dataset-specific rules. The quality rules Aa and Bb are mandatory, and the environmental rules Cc, Cd, De, and Df are optional.

S1.1 Overview and notation

Table S1 places each category in the taxonomy. Bb comprises six groups: common (Bb_1–Bb_6); uste16 (Bb_7); cic17 (Bb_8–Bb_12); cic18 (Bb_13–Bb_23); iot23 (Bb_24–Bb_26); Private #2 (Bb_27–Bb_29).

Level 1	Level 2 origin	Category (Level 3)	Rules	Status	Rule IDs
Data quality	A: collection environment	Aa : data quality	7	mandatory	Aa_1–Aa_7
Data quality	B: labeling	Bb : label errors and dataset-specific defects	29	mandatory	Bb_1–Bb_29
Environmental	C: network and host	Cc : network control	5	optional	Cc_1–Cc_5
Environmental	C: network and host	Cd : background	4	optional	Cd_1–Cd_4
Environmental	D: third parties	De : third-party CDN and SSO	4	optional	De_1–De_4
Environmental	D: third parties	Df : browser dependence	1	optional	Df_1
Total			50		

Table S1: Categories of the 50 refinement rules

Each rule is described in the form *rule ID: name—description (rationale), safeguard*, followed by its filter as an expression tagged with the rule ID. The filters use the following notation for a session: L_3 , L_4 , and L_7 are the network-, transport-, and application-layer protocols dissected by tshark; y is the class label and B denotes a benign label; *SYN*, *SA*, and *RST* indicate that a SYN, SYN-ACK, or RST segment is present, and FIN_c and FIN_s a FIN from the client or the server; *CH* and *SH* indicate a TLS ClientHello and ServerHello; W is the set of tshark expert-information messages; *sni*, *uri*, *ua*, and *q* are the TLS server name, HTTP URIs, HTTP user agent, and DNS query names; s and d are the source and destination addresses and P the set of the two ports; b , n , and τ are the payload bytes, packets, and duration of the session. $x \ni w$ means that x contains the string w , and $x \sim r$ that x matches the pattern r .

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S1.2 Aa: Data quality

Level 1: data quality noise · Level 2 origin: A: collection environment · 7 rules (Aa_1–Aa_7) · mandatory

Aa_1 Incomplete TCP 3-way handshake

Marks TCP sessions in which the handshake is incomplete, that is, in which a SYN or a SYN-ACK is missing. (A normal TCP connection always begins with a 3-way handshake [1]; its absence means that packets were lost during collection or removed by the publisher, so the session is noise for an NTC model that assumes a normal network environment.)

$$L_4 = \text{TCP} \wedge \neg(\text{SYN} \wedge \text{SA}) \quad (\text{Aa}_1)$$

Aa_2 Capture damage or truncation

Marks sessions that carry a tshark expert warning of capture damage: *packet size limited during capture*, *malformed packet*, or *truncated* [2]. (These warnings state that the capture itself is broken.) Safeguard: duplicate-ACK and previous-segment-not-captured warnings are excluded, because they are normal by-products of reassembly under high-rate floods; including them would have wrongly removed 95.9% of the PSH-ACK flood class of *iot23*.

$$W \cap \{\text{size-limited, malformed, truncated}\} \neq \emptyset \quad (\text{Aa}_2)$$

Aa_3 No FIN in either direction

Marks TCP sessions in which neither side sends a FIN. (A connection closed normally leaves a FIN from both sides [1].) Safeguard: a connection may also be aborted with RST or left open at the end of a capture, so the absence of FIN alone does not establish incomplete collection—66.3% of the *cispec* sessions, whose handshakes are intact, have no FIN. Aa_3 is therefore marked for information only and is applied to none of the ten datasets.

$$L_4 = \text{TCP} \wedge \neg \text{FIN}_c \wedge \neg \text{FIN}_s \quad (\text{Aa}_3)$$

Aa_4 Malformed packets

Marks sessions containing a packet that tshark cannot dissect. (A malformed packet is damaged or truncated data rather than a valid protocol message [2].)

$$W \ni \text{malformed} \quad (\text{Aa}_4)$$

Aa_5 Missing TLS Hello

Marks TLS sessions without a ClientHello or a ServerHello. (Every TLS session begins with this exchange [3], so its absence indicates collection damage.) Safeguard: the rule fires only for sessions that are actually TLS—application-layer TLS, port 443, or a Hello trace—so that non-TLS sessions are not removed, and it is not applied to datasets such as *tls1.3* whose publishers removed the handshake artificially.

$$\begin{aligned} L_4 &= \text{TCP} \wedge \text{TLS} \wedge (\neg \text{CH} \vee \neg \text{SH}), \\ \text{TLS} \equiv L_7 &= \text{tls} \vee 443 \in P \vee \text{CH} \vee \text{SH} \end{aligned} \quad (\text{Aa}_5)$$

Aa_6 Unlabeled or doubly labeled session

Marks sessions for which the label join yields no label (*unlabeled*) or conflicting labels (*double*). (A session without a unique label cannot serve as a training or test sample.)

$$y \in \{\text{unlabeled}, \text{double}\} \quad (\text{Aa_6})$$

Aa_7 IPv6 session

Marks sessions carried over IPv6 [4]. (Applications are diverse, but most application traffic in practice still runs over IPv4, whereas IPv6 is often used only inside the subnet, for link-local discovery, name resolution, and address configuration [5–7]. Such sessions carry subnet-internal control rather than the signal of an application or an attack and must be removed. In vpn16, 99.6% of the 122,835 IPv6 sessions are LLMNR, all addressed to multicast or link-local destinations.) In the application datasets Cc_1 also marks these sessions, whereas in the attack datasets Aa_7 alone removes them: 146,008 benign sessions of cic18, 955 sessions of Private #2, and 108 benign sessions of iot23 that no other rule marks.

$$L_3 = \text{IPv6} \quad (\text{Aa_7})$$

S1.3 Bb: Label errors and dataset-specific defects

Level 1: data quality noise · Level 2 origin: B: labeling · 29 rules (Bb_1–Bb_29) · mandatory

S1.3.1 Common rules (all datasets)

Bb_1 Infrastructure management protocols (common, class-aware)

Marks SNMP [8], NTP [9], RTCP [10], STUN [11], CHARGEN [12], and BitTorrent local service discovery (LSD) sessions that carry an application label. (They manage the network or the media path and are not produced by the labeled application.) Safeguard: STUN and RTCP in VoIP classes are kept, because they are part of the VoIP signal.

$$\begin{aligned} L_7 \in \mathcal{M} \wedge \neg(\text{VoIP}(y) \wedge L_7 \in \{\text{stun}, \text{rtcp}\}), \\ \mathcal{M} = \{\text{snmp}, \text{ntp}, \text{rtcp}, \text{lsd}, \text{chargin}, \text{stun}\} \end{aligned} \quad (\text{Bb_1})$$

Bb_2 P2P background (common, class-aware)

Marks BitTorrent DHT [13] and tracker sessions and sessions on ports 6969 and 61009 that carry a non-P2P label. (DHT and tracker maintenance runs regardless of the labeled application.) Safeguard: P2P classes keep these sessions as signal.

$$\begin{aligned} (L_7 \in \{\text{bt-dht}, \text{bt-tracker}\} \\ \vee P \cap \{6969, 61009\} \neq \emptyset) \wedge \neg \text{P2P}(y) \end{aligned} \quad (\text{Bb_2})$$

Bb_3 Class-protocol mismatch (common)

Marks sessions whose protocol contradicts the class name: an SCP class session that is not SSH [14], or an FTP [15] class session that is SIP [16]. (A label that contradicts the observed protocol is a labeling error.) Safeguard: network control protocols are not treated as mismatches, since they are handled by Cc; $\mathcal{C}$ is the set of control protocols of Cc_1–Cc_4.

$$\begin{aligned} [(y \ni \text{scp} \wedge L_7 \neq \text{ssh}) \\ \vee (y \ni \text{ftp} \wedge L_7 = \text{sip})] \wedge L_7 \notin \mathcal{C} \end{aligned} \quad (\text{Bb_3})$$

Bb_4 SNI-domain label mismatch (common)

Marks sessions whose TLS server name [17] is neither the labeled domain D nor one of its subdomains. (In a domain-labeled dataset, a session to another server is mislabeled.) Safeguard: the rule is restricted to sessions that have both an SNI and a domain label, that is, to domain-labeled datasets.

$$\text{sni} \neq \emptyset \wedge D \neq \emptyset \wedge \text{sni} \notin \{D\} \cup \{*.D\} \quad (\text{Bb_4})$$

Bb_5 Attack signature under a benign label (common)

Marks benign-labeled sessions that carry a high-precision attack fingerprint: a bot identifier, an uploaded file name, or an injection pattern (script, SQL, or path traversal) in the URI. (Such sessions are attacks hidden under a benign label.) Safeguard: low-precision cues—advertising and tracking URIs, TLS

heartbeats of normal CDNs, and ordinary SSH or FTP use—are deliberately excluded, since they would remove large amounts of benign traffic.

$$B \wedge (botid \vee upload \vee uri \sim \mathcal{P}_{inj}) \quad (\text{Bb_5})$$

Bb_6 Gateway VPN beacon in an attack label (common)

Marks attack-labeled single UDP packets that one host sends to port 7001 in the carrier-grade NAT range [18]. (They are health-check beacons of a gateway VPN that fell inside an attack window, not attacks.) Safeguard: the same beacons under a benign label are correctly labeled and are not touched.

$$\neg B \wedge L_4 = \text{UDP} \wedge d \in 100.64.0.0/10 \wedge p_d = 7001 \quad (\text{Bb_6})$$

S1.3.2 Dataset-specific rules: ustd16

Bb_7 Operating-system traffic of the infected VM (ustd16)

Marks malware-labeled sessions that are automatic traffic of the infected Windows XP virtual machine: name resolution and discovery (LLMNR [6], NetBIOS [19, 20], mDNS [21], SSDP, WS-Discovery), address assignment (DHCP [22], DHCPv6 [7]), layer-2 control (ARP [23], ICMPv6 [24]), and connectivity or proxy discovery (msftncsi, wpad, teredo, isatap, time.windows.com). (The operating system produced this traffic, but it inherited the malware label of the capture.) Safeguard: the same traffic in benign classes is correctly labeled and is kept; $\mathcal{O}$ and $\mathcal{O}_p$ are the protocols and ports listed above.

$$\begin{aligned} \neg B \wedge (L_7 \in \mathcal{O} \vee P \cap \mathcal{O}_p \neq \emptyset \\ \vee L_3 \in \{\text{arp}, \text{rarp}\} \vee L_4 = \text{ICMPv6} \\ \vee q \sim \mathcal{P}_{os} \vee uri \sim \mathcal{P}_{os}) \end{aligned} \quad (\text{Bb_7})$$

S1.3.3 Dataset-specific rules: cic17

Bb_8 Infiltration class (cic17)

Marks the whole Infiltration class. (Meterpreter remote-shell traces were confirmed in its payload, but the attack cannot be reproduced at the session level without payload fields, and the class has only 20 sessions, too few to contribute to learning.)

$$y \ni \text{infiltration} \quad (\text{Bb_8})$$

Bb_9 FTP-Patator fingerprint under a benign label (cic17)

Marks benign sessions whose FTP PASS argument has the dictionary-attack form “digits, a literal $\backslash t$, dictionary word.” (11,181 of 11,181 FTP-Patator PASS arguments have this form and 0% of benign ones do, so the fingerprint separates attack and benign completely.)

$$B \wedge ftp_arg \ni \backslash t \quad (\text{Bb_9})$$

Bb_10 SSH-Patator tool fingerprint under a benign label (cic17)

Marks benign sessions whose SSH identification string [14] names an automation library: paramiko, libssh2, ganymed, or dropbear. (99.9% of the SSH-Patator sessions use paramiko, whereas benign users use JSch or OpenSSH.)

$$B \wedge ssh_id \sim \{\text{paramiko}, \text{libssh2}, \text{ganymed}, \text{dropbear}\} \quad (\text{Bb_10})$$

Bb_11 HULK URI under a benign label (cic17)

Marks benign sessions whose HTTP request URI [25] has the HULK form $/\text{?KEY=VALUE}$, with key and value of 3 to 10 upper-case letters. (The HULK tool builds exactly this URI; the pattern matches 156,545 of 156,546 HULK sessions, and its 1,916 benign hits equal the figure confirmed by manual review.)

$$B \wedge uri \sim \text{?}[A-Z]\{3,10\}=[A-Z]\{3,10\} \quad (\text{Bb}_{11})$$

Bb_12 GoldenEye URI under a benign label (cic17)

Marks benign sessions whose URI is the root path followed by one to five random parameters, not matched by Bb_11. (This is the request form of GoldenEye; it matches 7,031 of 7,379 GoldenEye sessions, 95.3%, and all 181 benign hits fall within the official attack window.) Safeguard: each key must contain an upper-case letter or a digit; benign parameter names are lower-case words, and without this condition 144 benign sessions would also match.

$$B \wedge \neg \text{Bb}_{11} \wedge uri \sim \mathcal{P}_{\text{GE}} \quad (\text{Bb}_{12})$$

S1.3.4 Dataset-specific rules: cic18**Bb_13 Attacker address under a benign label (cic18)**

Marks benign sessions involving one of the 23 attacker addresses $\mathcal{A}$ of the official attack schedule [26]. (The labels were assigned by attacker, victim, and time window; the official windows are narrower than the actual attacks, so attack sessions outside them were labeled benign.) Safeguard: these addresses belong to dedicated attack machines, on which benign traffic cannot exist.

$$B \wedge (s \in \mathcal{A} \vee d \in \mathcal{A}) \quad (\text{Bb}_{13})$$

Bb_14 Reconnaissance and scanner traffic under a benign label (cic18)

Marks benign sessions with byte-level scanner fingerprints: DNS server-identification queries (version.bind, hostname.bind, id.server [27]), scanner user agents (zgrab, masscan, nmap, nikto, sqlmap, dirbuster, gobuster, hydra, nuclei), and SIP scanner agents (friendly-scanner, sipvicious). (These are reconnaissance traffic, not benign use.)

$$B \wedge (q \sim \mathcal{P}_{\text{id}} \vee ua \sim \mathcal{P}_{\text{scan}} \vee sip_ua \sim \mathcal{P}_{\text{sip}}) \quad (\text{Bb}_{14})$$

Bb_15 FTP-Patator without an attack payload (cic18)

Marks the whole FTP-Patator class. (Port 21 was closed during the attack: sessions end in SYN followed by RST, and no USER or PASS command was ever sent, so no attack payload exists [28].)

$$y \ni \text{ftp-patator} \quad (\text{Bb}_{15})$$

Bb_16 Slowhttptest without an attack payload (cic18)

Marks the whole Slowhttptest class. (The attack was misdirected to port 21 and no successful attack traffic exists [28].)

$$y \ni \text{slowhttptest} \quad (\text{Bb}_{16})$$

Bb_17 SSH-Patator misfire to port 21 (cic18)

Marks SSH-Patator sessions sent to port 21. (They are misfires rather than SSH attacks; Liu *et al.* report the same 30 flows [28].)

$$y \ni \text{ssh-patator} \wedge p_d = 21 \quad (\text{Bb_17})$$

Bb_18 LOIC-UDP processed as ICMP (cic18)

Marks LOIC-UDP sessions whose transport protocol is ICMP [29]. (An old CICFlowMeter bug processed these flows incorrectly [28].)

$$y \ni \text{loic-udp} \wedge L_4 = \text{ICMP} \quad (\text{Bb_18})$$

Bb_19 Empty Slowloris flow (cic18)

Marks Slowloris sessions that carry no payload. (The 120-second timeout of the flow extractor split slow connections into fragments without payload [28].)

$$y \ni \text{slowloris} \wedge b = 0 \quad (\text{Bb_19})$$

Bb_20 Early-reset GoldenEye attempt (cic18)

Marks GoldenEye sessions that end with RST within 5.05 s. (They are incomplete attempts rejected under overload, an artifact rather than attack traffic [28].)

$$y \ni \text{goldeneye} \wedge RST \wedge \tau < 5.05 \quad (\text{Bb_20})$$

Bb_21 Page resources in web attacks (cic18)

Marks web-attack sessions that request page resources (CSS, JavaScript, images, icons, fonts). (They are side requests of the attacked page, not the attack itself.) $\mathcal{P}_{\text{res}}$ matches URIs ending in css, js, png, jpg, jpeg, gif, ico, woff, woff2, or svg.

$$y \ni \text{web-attack} \wedge \text{uri} \sim \mathcal{P}_{\text{res}} \quad (\text{Bb_21})$$

Bb_22 Empty Bot flow (cic18)

Marks Bot sessions without payload. (TCP segment offsets in flow extraction created payload-free flows [28].)

$$y \ni \text{bot} \wedge b = 0 \quad (\text{Bb_22})$$

Bb_23 Internal NMAP scan under a benign label (cic18)

Marks benign SYN-only sessions from the two infected victims (172.31.69.24 and 172.31.69.13) to internal hosts. (After infiltration, the victims scanned the internal network with NMAP, and these scans were labeled benign [28].)

$$\begin{aligned} B \wedge s \in \{172.31.69.24, 172.31.69.13\} \\ \wedge d \in 172.31.69.0/24 \wedge SYN \wedge \neg SA \end{aligned} \quad (\text{Bb_23})$$

S1.3.5 Dataset-specific rules: **iot23**

Bb_24 Session with an Internet endpoint in an attack capture (iot23)

Marks attack-labeled sessions with a public address [30] at either end. (All attacks are performed by malicious IoT devices against IoT devices inside the testbed, while the testbed is connected to the Internet and each capture is labeled as a whole with its attack [31]; sessions with the Internet cannot be the attack.) Safeguard: the public range of the dataset creators (205.174.165.0/24), whose role is not documented, is conservatively excluded.

$$\neg B \wedge (\text{pub}(s) \vee \text{pub}(d)) \wedge \neg \text{cic}(s, d) \quad (\text{Bb_24})$$

Bb_25 Internal infrastructure protocols in an attack capture (iot23)

Marks attack-labeled internal sessions of discovery and infrastructure protocols $\mathcal{I}$ that none of the 33 attacks uses: STP, CDP, LLDP, IGMP, ICMPv6, RARP, DHCP, DHCPv6, mDNS, LLMNR, NetBIOS, SSDP, WS-Discovery, CoAP, NTP, and local service discovery. (They are the constant background of the devices [31].) Safeguard: ARP and DNS are deliberately excluded, because they are the vehicles of the ARP-spoofing, DNS-spoofing, and UDP-flood attacks; including them would remove attack traffic itself.

$$\begin{aligned} &\neg B \wedge \neg \text{pub}(s) \wedge \neg \text{pub}(d) \wedge L_7 \in \mathcal{I} \\ &\wedge \neg \text{Bb_26} \wedge \neg \text{cic}(s, d) \end{aligned} \quad (\text{Bb_25})$$

Bb_26 Broadcast, multicast, or link-local destination in an attack capture (iot23)

Marks attack-labeled sessions to a broadcast [32], multicast [5, 33], or link-local [5, 34] destination. (A targeted attack cannot be addressed to such a destination.) Safeguard: for encapsulated sessions only the outer destination is checked; checking the spoofed inner header of GRE floods would have produced 9,941 false marks in Mirai-Greeth and 4,637 in Mirai-Greip.

$$\neg B \wedge \text{bcast}(d_{\text{outer}}) \wedge \neg \text{cic}(s, d) \quad (\text{Bb_26})$$

S1.3.6 Dataset-specific rules: **Private #2**

Bb_27 nmap UDP probe under a benign label (Private #2)

Marks benign sessions whose UDP payload matches a probe of the nmap-payloads database [35]. (They are scans hidden in benign captures.) Safeguard: the byte-level match was cross-checked against the manual review (e.g., the 835 version.bind probes of the UCloud class).

$$B \wedge \text{nmap} \quad (\text{Bb_27})$$

Bb_28 SYN-only scan in the UCloud class (Private #2)

Marks single-packet SYN-only TCP sessions in the benign UCloud class. (Unanswered single SYNs are port scans, not cloud-service use.) Safeguard: the class condition confines the rule to this class, so no attack class is affected.

$$\begin{aligned} &y = \text{UCloud} \wedge L_4 = \text{TCP} \\ &\wedge \text{SYN} \wedge \neg \text{SA} \wedge n = 1 \end{aligned} \quad (\text{Bb_28})$$

Bb_29 SYN-only sweep in the TencentCloud class (Private #2)

Marks single-packet SYN-only sessions in the benign TencentCloud class, which sweep the target ports of IoT botnets (23, 2323, 37215, 445, and others) from the Tencent address range. (They are port scans, not cloud-service use.)

$$y = \text{TencentCloud} \wedge \text{SYN} \wedge \neg \text{SA} \wedge n = 1 \quad (\text{Bb_29})$$

S1.4 Cc: Network control

Level 1: environmental noise · Level 2 origin: C: network and host · 5 rules (Cc_1–Cc_5) · optional

Cc_1 Layer-3 control and IPv6

Marks ARP [23], RARP, ICMP [29], ICMPv6 [24], and Teredo [36] sessions and all IPv6 [4] sessions. (Layer-3 control is common to every application and identifies none of them; IPv6 sessions are also marked by Aa_7.)

$$\begin{aligned} L_7 &\in \{\text{arp}, \text{rarp}, \text{icmp}, \text{icmpv6}, \text{teredo}\} \\ \vee L_3 &\in \{\text{IPv6}, \text{arp}, \text{rarp}\} \vee L_4 \in \{\text{icmp}, \text{icmpv6}\} \end{aligned} \quad (\text{Cc}_1)$$

Cc_2 Name resolution and address assignment

Marks DNS [37], mDNS [21], LLMNR [6], NBNS [19, 20], DHCP [22], DHCPv6 [7], and BOOTP sessions. (Every application triggers them, so they identify no application.)

$$\begin{aligned} L_7 &\in \{\text{dns}, \text{mdns}, \text{llmnr}, \text{nbns}, \\ &\quad \text{dhcp}, \text{dhcpv6}, \text{bootp}\} \end{aligned} \quad (\text{Cc}_2)$$

Cc_3 Windows network services

Marks sessions of the NetBIOS datagram and session services [19, 20], SMB, SMB2, the Service Location Protocol [38], and SSDP. (They are operating-system service traffic, not application traffic.)

$$L_7 \in \{\text{nbdgm}, \text{nbss}, \text{smb}, \text{smb2}, \text{srvloc}, \text{ssdp}\} \quad (\text{Cc}_3)$$

Cc_4 Infrastructure management

Marks SNMP [8], NTP [9], RTCP [10], STUN [11], CHARGEN [12], and LSD sessions. (Management, time synchronization, and media control are independent of the task.) Safeguard: Cc_4 itself has no class exception; datasets in which STUN and RTCP are signal, such as VoIP, keep them through the class-aware rule Bb_1.

$$L_7 \in \mathcal{M} \quad (\text{Cc}_4)$$

Cc_5 Broadcast, multicast, and link-local destinations

Marks sessions to a broadcast address [32], an IPv4 or IPv6 multicast address [5, 33], or a link-local address [5, 34]. (A session without a specific peer is not the traffic of a targeted application.)

$$\begin{aligned} d &\in \{255.255.255.255, *.255\} \vee d \in 224.0.0.0/4 \\ \vee d &\in 169.254.0.0/16 \vee d \in \text{ff00::}/8 \\ \vee d &\in \text{fe80::}/10 \end{aligned} \quad (\text{Cc}_5)$$

S1.5 Cd: Background

Level 1: environmental noise · Level 2 origin: C: network and host · 4 rules (Cd_1–Cd_4) · optional

Cd_1 P2P DHT and tracker maintenance

Marks BitTorrent DHT [13] and tracker sessions and sessions on ports 6969 and 61009. (DHT and tracker maintenance runs continuously in the background, independent of the application in use.) Safeguard: BitTorrent classes, for which this traffic is signal, keep it through the class-aware rule Bb_2.

$$L_7 \in \{\text{bt-dht}, \text{bt-tracker}\} \vee P \cap \{6969, 61009\} \neq \emptyset \quad (\text{Cd}_1)$$

Cd_2 Windows background traffic

Marks Windows connectivity probes (msftncsi in DNS queries or URIs), sessions with the winhttp user agent, and Windows telemetry and runtime-update servers $\mathcal{W}$. (They are automatic background traffic of the operating system, not user activity.) Safeguard: in a DNS stream a C2 query of a domain-generation algorithm can occasionally (about 3%) share a session with a connectivity probe, so the rule should be used with care in DGA-detection datasets.

$$\begin{aligned} ua &= \text{winhttp} \vee sni \in \mathcal{W} \\ \vee q \ni \text{msftncsi} \vee uri \ni \text{msftncsi} \end{aligned} \quad (\text{Cd}_2)$$

Cd_3 Microsoft update delivery

Marks sessions to the Microsoft delivery-optimization and Windows Update servers (*.do.dsp.mp.microsoft.com, dl.delivery.mp.microsoft.com, and download.windowsupdate.com). (Automatic downloads of the operating system and store are not application signal.)

$$sni \sim \mathcal{P}_{\text{MS}} \quad (\text{Cd}_3)$$

Cd_4 Package-manager updates

Marks HTTP sessions of the Linux package manager (user agent apt-http). (Automatic package updates are not user applications.)

$$ua \ni \text{apt-http} \quad (\text{Cd}_4)$$

S1.6 De: Third-party CDN and SSO

Level 1: environmental noise · Level 2 origin: D: third parties · 4 rules (De_1–De_4) · optional

De_1 Google shared infrastructure and SSO

Marks TLS and gQUIC sessions to Google static and authentication servers (server names containing gstatic or google). (Many applications share these servers, so they do not identify an application.) Safeguard: content servers (googlevideo) are excluded, and class-aware exceptions keep legitimate cues for Google application classes.

$$\begin{aligned} & (sni \ni \text{gstatic} \vee sni \ni \text{google}) \\ & \wedge sni \not\ni \text{googlevideo} \wedge L_7 \in \{\text{tls}, \text{gquic}\} \end{aligned} \quad (\text{De}_1)$$

De_2 Mail SSO and CDN

Marks sessions to mail servers shared by many services (server names containing mail). (Shared mail infrastructure carries no signal of a particular application.)

$$sni \ni \text{mail} \quad (\text{De}_2)$$

De_3 Advertising and tracking

Marks sessions to third-party advertising and analytics servers (e.g., doubleclick.net, googlesyndication.com, google-analytics.com, and adnxs.com). (Advertising and analytics are inserted independently of the content.)

$$sni \sim \mathcal{P}_{\text{ad}} \quad (\text{De}_3)$$

De_4 Browser infrastructure

Marks sessions to browser safety, telemetry, and shared static servers (safebrowsing.google.com, clientsN.google.com, *.gstatic.com, and *.fbcdn.net). (Browsers call them automatically, not the user.)

$$sni \sim \mathcal{P}_{\text{br}} \quad (\text{De}_4)$$

S1.7 Df: Browser dependence

■ *Level 1: environmental noise* · *Level 2 origin: D: third parties* · *1 rule (Df_1)* · *optional*

Df_1 Browser start page

Marks requests to the default start and search pages that browsers load automatically (URIs containing msn.com or bing). (This traffic is caused by the browser, not by the application under study.)

$$uri \ni \text{msn.com} \vee uri \ni \text{bing} \quad (\text{Df_1})$$

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